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Dynamic Programming for Edit Distance and Optimal Substructures

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Domain	DOM_CS

Abstract

The string-to-string correction problem requires computing the minimum cost sequence of insertions, deletions, and substitutions. We provide an O(mn) algorithm exploiting optimal subproblems in matrix recurrence.

Abstract quoted from R. A. Wagner et al., Journal of the ACM (1974).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?